ABOUT UNIX

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->UNIX is an operating system developed at Bell Laboratories in 1969. Ken Thompson and Dennis Ritchie were the key contributors to its development.

->UNIX was written in assembly language. After the development of the C programming language, UNIX was largely rewritten in C in 1973.

Why was UNIX rewritten in C?

The main reason was portability. Assembly language Code written for one processor may not work on another processor because their instructions can differ.

What is portability?

Portability means that software can be adapted to run on different hardware platforms with fewer changes to its source code.

Disadvantages of UNIX:

1. Difficult to learn: Beginners may find UNIX commands and concepts difficult to understand.

2. Command-line interface: Many tasks require commands, which can be challenging for users who prefer a graphical interface.

3. Cost: Some commercial UNIX operating systems require paid licenses.

4. Software compatibility: Some applications developed for one UNIX version may not work on another version.

5. Hardware compatibility: Certain UNIX versions support only specific hardware platforms.

6. Technical knowledge: Installation, configuration, and troubleshooting may require specialized skills.

ABOUT LINUX:

->LINUX is an operating system Created by the Linus Torvalds in 1991.

->Linux is mainly written in C language, with some parts written in Assembly Language.

Why some parts written in Assembly Language:

->Linux is mainly Written in c but some parts are written in Assembly Language because Assembly provides direct access to processor-specific instructions and hardware Operations.

Why do we use Assembly in Linux?

1. CPU control: Assembly allows direct use of processor instructions.

2. Hardware initialization: Some low-level operations are needed when the computer starts.

3. Interrupt handling: Some processor-specific interrupt and exception operations require low-level code.

4. Context switching: Switching CPU execution from one process or thread to another can require architecture-specific assembly instructions.

5. Special CPU instructions: Some processor operations are easier or only directly accessible through Assembly or compiler-specific interfaces.

Why not write everything in C?

C is portable, but it does not directly express every processor-specific operation needed by an operating system.

Assembly provides that low-level control, although modern compilers can generate assembly instructions from C code when appropriate.

ALREADY UNIX IS THERE WHY WE USE LINUX?

UNIX is not open source and paid Licenses, that's why we go for Linux. Linux is open-source, free of cost, easy to modify the code**.**

Where is Linux used today?

- Servers: Websites and online applications.

- Cloud computing: Cloud infrastructure and services.

- Android phones: Android uses the Linux kernel.

- Embedded systems: Routers, smart devices, and industrial equipment.

- Software development: Programmers use Linux to develop and test applications.